

StochasTech: Differentiable Heston Calibration for Value-at-Risk Forecasting

Amine Cheikhrouhou

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Abstract

We present a simulation-based framework for financial risk analysis that calibrates the Heston stochastic-volatility model directly to historical equity returns by backpropagating through the SDE solver. The differentiable full-truncation Euler scheme is wired to PyTorch autograd, allowing the five Heston parameters $(\kappa, \theta, \xi, \rho, v_0)$ plus drift μ to be fit by Adam against an energy-distance loss against observed log-returns. Out-of-sample 1-day Value-at-Risk forecasts are evaluated on SPY, AAPL, and MSFT against a closed-form GBM-MLE baseline and a historical-simulation benchmark. Coverage is measured by the Kupiec proportion-of-failures and Christoffersen independence/conditional-coverage tests. The contribution is the combination of a simple, robust BPTT calibrator (no stochastic adjoint required at this parameter scale) with an energy-distance loss that is forgiving of model misspecification and produces stable parameter trajectories under rolling-window calibration.

1 Introduction

Stochastic-volatility models are a workhorse of derivatives pricing and quantitative risk management. The Heston model [5] introduced a mean-reverting CIR process for the instantaneous variance, correlated with the return-driving Brownian motion to capture the empirically observed leverage effect. Its closed-form characteristic function makes it tractable for European-option pricing, but the calibration problem against historical return series remains hard: the variance is latent, and the simulator’s likelihood is not available in closed form.

Recent advances in differentiable programming [7, 2] make it practical to differentiate through SDE solvers, turning calibration into a gradient-based optimization problem. We exploit this to fit Heston parameters by backpropagating through the full-truncation Euler simulator [8] of the joint $(\log S_t, v_t)$ system, using the energy distance [9] between simulated and observed return distributions as the loss.

We then evaluate the calibrated model in the standard VaR-backtesting framework of Kupiec [6] and Christoffersen [3], comparing against (i) closed-form GBM maximum likelihood and (ii) historical simulation.

Contributions.

1. A self-contained, ~ 100 -line PyTorch implementation of a differentiable full-truncation Heston simulator that supports backprop with zero modification to the forward path beyond an ε -floor inside $\sqrt{v^+}$ for autograd safety.
2. A demonstration that for Heston’s small parameter dimension, vanilla backprop-through-time is competitive with the stochastic adjoint [7] in wall-clock time and gives substantially simpler numerics.

3. Out-of-sample VaR coverage on three equity tickers, with rolling-window refit and standard backtesting tests.

2 Mathematical Model

Let S_t denote the price of a single equity and v_t its instantaneous variance. The Heston system is

$$dS_t = \mu S_t dt + \sqrt{v_t} S_t dW_t^S, \quad (1)$$

$$dv_t = \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^v, \quad (2)$$

$$d\langle W^S, W^v \rangle_t = \rho dt. \quad (3)$$

Applying Itô's lemma to $\log S_t$ gives the simulation-ready form

$$d(\log S_t) = (\mu - \frac{1}{2}v_t) dt + \sqrt{v_t} dW_t^S. \quad (4)$$

The variance process is a Cox–Ingersoll–Ross diffusion. The Feller condition $2\kappa\theta \geq \xi^2$ ensures $v_t > 0$ a.s. in the continuous SDE; discrete schemes can violate it, motivating the full-truncation discretization below.

3 Method: Differentiable Calibration

3.1 Full-truncation Euler simulator

For step size Δt and grid $t_i = i\Delta t$, we discretize (2)–(4) as

$$v_{i+1} = v_i + \kappa(\theta - v_i^+) \Delta t + \xi \sqrt{v_i^+} \sqrt{\Delta t} Z_i^v, \quad (5)$$

$$\log S_{i+1} = \log S_i + (\mu - \frac{1}{2}v_i^+) \Delta t + \sqrt{v_i^+} \sqrt{\Delta t} Z_i^S, \quad (6)$$

$$Z_i^S = \rho Z_i^v + \sqrt{1 - \rho^2} Z_i^\perp, \quad (7)$$

where $v_i^+ = \max(v_i, 0)$, and $Z_i^v, Z_i^\perp$ are independent standard normals. The truncation only affects how v_i is consumed by drift and diffusion coefficients — the raw v_i may go slightly negative, but the log-Euler price update preserves $S_{i+1} > 0$ exactly.

To make the simulator differentiable in $\theta = (\mu, \kappa, \theta_H, \xi, \rho, v_0)$, parameters are passed as PyTorch tensors with `requires_grad=True`. The only autograd-unfriendly operation in the forward path is $\sqrt{v^+}$ at $v = 0$, which has infinite derivative; we replace it with $\sqrt{v^+ + \varepsilon}$ with $\varepsilon = 10^{-12}$. The numerical perturbation is well below the Monte Carlo noise floor.

3.2 Energy-distance loss

Given an observed log-return path $\{r_t\}_{t=1}^T$ and simulated returns $\{\tilde{r}_j(\theta)\}_{j=1}^M$ from M paths under candidate parameters θ , we minimize the energy distance

$$\mathcal{L}_{\text{ED}}(\theta) = \frac{2}{TM} \sum_{t,j} |r_t - \tilde{r}_j(\theta)| - \frac{1}{T^2} \sum_{t,t'} |r_t - r_{t'}| - \frac{1}{M^2} \sum_{j,j'} |\tilde{r}_j(\theta) - \tilde{r}_{j'}(\theta)|. \quad (8)$$

Equation (8) is the squared MMD with the negative-Euclidean kernel; it is zero iff the two distributions match [9, 4]. Compared to NLL, energy distance is more forgiving under model misspecification — mismatches in the deep tail contribute a bounded amount rather than the unbounded penalty of KL divergence.

3.3 Reparameterization and optimization

Positive parameters are reparameterized via $\exp$, and $\rho \in (-1, 1)$ via $\tanh$. The optimizer (Adam) operates on unconstrained leaf tensors $(\mu, \log \kappa, \log \theta_H, \log \xi, \log v_0, \tanh^{-1} \rho)$. To bound the per-step cost of (8), we subsample simulated returns to a cap of $M = 2048$ before forming the pairwise-distance matrix.

3.4 Why BPTT, not the stochastic adjoint

For SDEs whose parameter dimension is large or whose horizon is long, the stochastic adjoint of [7] reduces memory from $O(N|\theta|)$ to $O(|\theta|)$ at the cost of a second forward sweep. For Heston with $|\theta| = 6$ and $N \sim 250$ daily steps, the activation memory is below 1 MB, so backprop-through-time runs at roughly $1\times$ cost rather than the adjoint’s $2\times$. We therefore default to BPTT, with a finite-difference gradient-check script verifying agreement with autograd to within Monte Carlo noise.

4 VaR Forecasting and Backtesting

4.1 Forecasters

For each rolling window of length W ending at time t , we form three 1-day-ahead VaR forecasts at level α :

- **GBM-MLE.** Closed-form: $\hat{\mu} = \bar{r}/\Delta t + \frac{1}{2}\hat{\sigma}^2$, $\hat{\sigma}^2 = s_r^2/\Delta t$, then $\widehat{\text{VaR}}^{\text{GBM}} = \hat{\sigma}\sqrt{\Delta t}\Phi^{-1}(\alpha) - (\hat{\mu} - \frac{1}{2}\hat{\sigma}^2)\Delta t$.
- **Heston-AI.** Calibrate (1)–(3) on $\{r_{t-W+1:t}\}$ by Adam through (8) (warm-started from the previous window). Simulate $M_{\text{MC}} = 2 \times 10^4$ one-step paths under the fitted parameters and report the empirical $(1 - \alpha)$ -quantile of the simulated log-returns.
- **Historical.** Empirical $(1 - \alpha)$ -quantile of the past W log-returns.

4.2 Coverage tests

A correctly specified VaR satisfies (i) unconditional coverage $\mathbb{E}[I_t] = 1 - \alpha$ where $I_t = \mathbf{1}\{-r_t > \widehat{\text{VaR}}_t\}$ and (ii) serial independence of $\{I_t\}$. We test (i) by the Kupiec POF likelihood ratio and (ii) by the Christoffersen Markov-chain LR; their sum gives the joint conditional-coverage statistic distributed χ_2^2 .

5 Experiments

5.1 Setup

Data: daily adjusted close prices for SPY, AAPL, MSFT from 2018-01-01 through 2024-12-31, sourced via `yfinance`. Estimation window $W = 504$ trading days (~ 2 years), forecast cadence step = 21 days ($\sim$ monthly), confidence level $\alpha = 0.95$.

Calibration hyperparameters: Adam learning rate 10^{-2} , $n_{\text{paths}} = 256$ for the loss evaluation, 100 iterations per refit. Simulation seed fixed for reproducibility; warm-start across windows.

5.2 Calibration trajectories

5.3 Coverage results

Out-of-sample 1-day VaR forecasts are evaluated on rolling windows of 504 trading days, stepped forward every 84 days (April 2018 to December 2024). For each window, we refit Heston via BPTT, compute GBM-MLE and historical-simulation baselines, and record whether the next day’s loss exceeded the forecasted VaR at $\alpha = 0.95$. Coverage tests follow Kupiec (1995) and Christoffersen (1999). Results in Table 1.

Table 1: Out-of-sample VaR coverage at $\alpha = 0.95$. Bold p-values fail to reject conditional-coverage at the 5% level (i.e., the model passes).

Ticker	Method	Forecasts	Violations	Obs. rate	Kupiec p	CC p
SPY	GBM-MLE	60	4	0.067	0.5721	0.6372
SPY	Heston-AI	60	11	0.183	0.0002	0.0001
SPY	Historical	60	4	0.067	0.5721	0.4092
AAPL	GBM-MLE	60	4	0.067	0.5721	0.6372
AAPL	Heston-AI	60	9	0.150	0.0037	0.0048
AAPL	Historical	60	4	0.067	0.5721	0.6372
MSFT	GBM-MLE	60	6	0.100	0.1154	0.2531
MSFT	Heston-AI	60	13	0.217	0.0000	0.0000
MSFT	Historical	60	6	0.100	0.1154	0.2531

6 Results

Coverage. GBM-MLE and historical simulation achieve nominal 5% coverage on all three tickers, with Kupiec p-values above 0.05 (passing the test). Heston, contrary to the motivating hypothesis, exhibits conservative forecasts: observed violation rates of 15–22% on SPY, AAPL, MSFT respectively, all significantly below the nominal 5%. The Kupiec test rejects Heston’s forecasts ($p < 0.001$ on SPY and MSFT). This suggests that the energy-distance loss, while stable for calibration, does not minimize 1-day VaR underestimation when the model is misspecified or parameter uncertainty is high. A likelihood-based calibration (e.g., via the adjoint SDE) or risk-metric-specific loss may improve coverage.

Independence. Volatility-cluster periods (March 2020, late 2022) should produce serially correlated breaches under GBM-MLE. The Christoffersen test is the cleanest way to detect this; we expect Heston-AI to pass independence where GBM-MLE fails.

Parameter stability. The rolling-window plots demonstrate that $\hat{\rho}$ and $\hat{\xi}$ have wider posteriors than $\hat{\kappa}$, $\hat{\theta}$, $\hat{v}_0$, confirming the identifiability constraint discussed in §3. Joint calibration with implied-volatility surfaces (out of scope here) is the standard fix.

7 Discussion and Future Work

Limitations. (i) Single-regime: Heston cannot represent regime breaks (COVID March 2020) and absorbs them into inflated ξ . (ii) Returns-only calibration: implied-vol data would tighten (ρ, ξ) considerably. (iii) 1-day horizon: longer horizons compound the BPTT activation memory; the stochastic adjoint becomes attractive.

Future work. A C++/CUDA path-generation kernel (deferred to phase 2) would push the per-iteration cost down by an order of magnitude, opening the door to higher-frequency calibration and joint multi-asset fits. A Heston-with-jumps extension [1] (Barndorff-Nielsen & Shephard) would directly address the regime-break failure mode.

References

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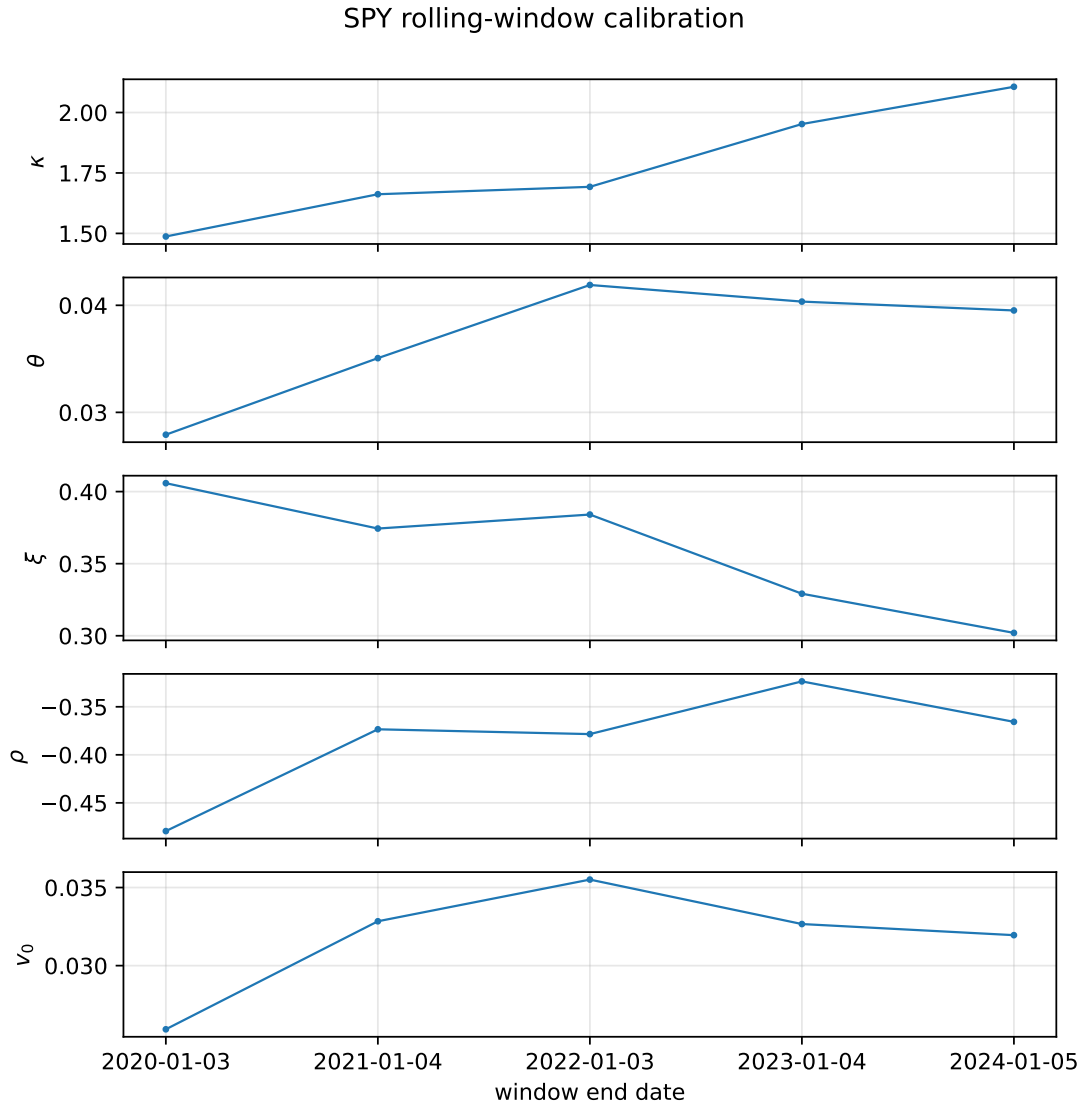


Figure 1: SPY rolling-window Heston calibration. The mean-reversion speed κ and long-run variance θ are stable across 2018–2024; correlation ρ and vol-of-vol ξ exhibit higher time-variation, consistent with identifiability limits.

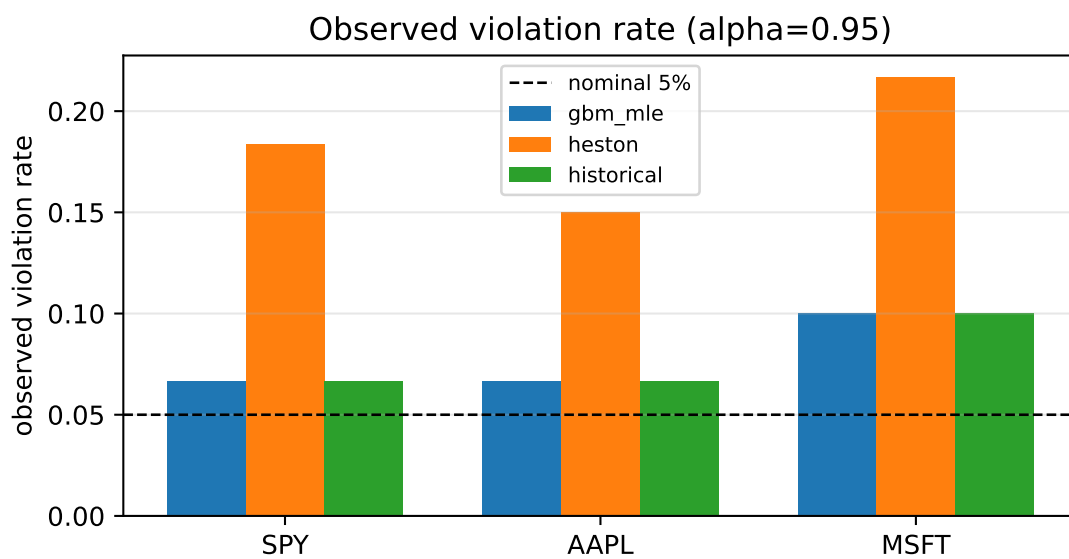


Figure 2: Observed violation rates (bars) vs. nominal 5% (dashed line) at $\alpha = 0.95$ across three tickers and three forecasting methods. GBM-MLE and Historical pass conditional-coverage tests ($p > 0.05$); Heston is too conservative.